Dynamical Origins of Speed–Accuracy Trade-offs and Dysfunctional Decision Regimes in a Reduced Cortical Decision Circuit

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Bachelor's Project Report

Abstract

I study perceptual decision-making in the reduced two-variable cortical circuit of Wong and Wang (2006) using simulation and phase-plane analysis only. Two populations with slow NMDA-mediated recurrent excitation compete through cross-inhibition and integrate noisy evidence. After validating baseline behavior (psychometric and chronometric curves, slower errors), I map the deterministic geometry of the circuit (nullclines, saddle, separatrix, bifurcations) and the stochastic behavior over the plane of cross-inhibition J_{12} and noise amplitude σ (15×15 cells, 500 trials each). Two hypotheses are tested. **H1:** reaction time (RT) and error rate are explained by the saddle's unstable eigenvalue λ_u, the distance d of the resting state to the separatrix, and σ. This is *partly* supported: λ_u and σ explain $R^2 = 0.78$ of log RT, whereas d is nearly constant and uninformative, and the error rate depends on σ alone. **H2:** behavioral regime boundaries coincide with deterministic bifurcation curves shifted by noise. This is *not supported* in its strong form: the noise-induced impulsive boundary is predicted by a deterministic barrier proxy ($R^2 = 0.98$) but not by bifurcation values; an indecisive regime appears only when recurrent excitation J_{11} is weakened, and its boundary follows the point where the attractor rate equals the decision criterion rather than the existence bifurcation. An empirical $-\log P$ landscape shows the double-well structure collapsing at $\sigma \approx 0.06$ nA, matching the noise level at which accuracy breaks down.

1 Introduction

In random-dot motion tasks, neurons in the lateral intraparietal area ramp their firing rates in a way consistent with accumulation of evidence to a threshold (Roitman and Shadlen, 2002). Attractor-network models explain this ramping as a transition from a spontaneous state, through the vicinity of a saddle point, into one of two choice attractors (Wang, 2002; Wong and Wang, 2006). The reduced two-variable model can be approximated by a one-dimensional diffusion equation (Roxin and Ledberg, 2008) and is related to drift–diffusion models (Bogacz et al., 2006).

Research question. How do the ratio of recurrent excitation to cross-inhibition (J_{11}/J_{12}) and the background noise amplitude σ shape the phase-plane geometry of the decision circuit, and to what extent do deterministic geometric quantities predict stochastic behavior (accuracy, reaction time, regime)? The model has no explicit inhibitory population, so J_{11}/J_{12} is an *effective* ratio of self-excitation to cross-inhibition, not an E/I ratio.

Hypotheses. **H1.** Across the (J_{12}, σ) plane, mean RT and error rate at fixed coherence are explained by the unstable eigenvalue λ_u of the saddle, the distance d from the start state to the separatrix, and σ. **H2.** Boundaries between behavioral regimes (normal, impulsive, indecisive) coincide with deterministic bifurcation curves, shifted by noise in a predictable direction. Negative and partial results are reported as such.

2 Model

Each population $i \in \{1, 2\}$ has an NMDA gating variable S_i:

$$\frac{dS_i}{dt} = -\frac{S_i}{\tau_S} + (1 - S_i)\,\gamma\,H(x_i), \qquad H(x) = \frac{ax - b}{1 - \exp[-d(ax - b)]}, \tag{1}$$

$$x_1 = J_{11}S_1 - J_{12}S_2 + I_0 + J_{\text{ext}}\mu_0\big(1 + \tfrac{c'}{100}\big) + I_{\text{noise},1}, \tag{2}$$

$$x_2 = J_{11}S_2 - J_{12}S_1 + I_0 + J_{\text{ext}}\mu_0\big(1 - \tfrac{c'}{100}\big) + I_{\text{noise},2}, \tag{3}$$

where c' is the signed coherence (%) favoring population 1. Baseline parameters follow Wong and Wang (2006): $\tau_S = 100$ ms, $\gamma = 0.641$, $a = 270$ Hz/nA, $b = 108$ Hz, $d = 0.154$ s, $J_{11} = 0.2609$ nA, $J_{12} = 0.0497$ nA, $I_0 = 0.3255$ nA, $\mu_0 = 30$ Hz, $J_{\text{ext}} = 5.2 \times 10^{-4}$ nA/Hz (Appendix A). Noise is an Ornstein–Uhlenbeck current,

$$\tau_{\text{AMPA}}\,dI_{\text{noise,i}} = -I_{\text{noise,i}}\,dt + \sqrt{\tau_{\text{AMPA}}}\,\sigma\,dW_i, \qquad \tau_{\text{AMPA}} = 2 \text{ ms}, \tag{4}$$

with stationary standard deviation $\sigma/\sqrt{2}$ and baseline $\sigma = 0.02$ nA. Parameters should be verified against the original publication before reuse; I could not check them against the paper text and relied on the values given in the project proposal, which reproduce the published qualitative behavior (Section 4.1).

3 Methods

Simulation. Trials start at the deterministic resting state (stimulus absent) and run 500 ms of pre-stimulus noise, then the stimulus for up to 2 s. The gating variables are advanced with the Euler scheme ($dt = 0.5$ ms), and the OU currents with their *exact* Gaussian transition (an Euler–Maruyama step for the OU process with $dt/\tau_{\text{AMPA}} = 0.25$ overestimates its variance by about 14%). Code is compiled with Numba; trial i uses seed $s_0 + i$, so every number is reproducible. Convergence is checked against $dt \in \{1, 0.5, 0.25, 0.1\}$ ms.

Decision criterion. The read-out is $r_i = S_i/(\gamma\tau_S(1 - S_i))$, which equals $H(x^*)$ at any fixed point but is smooth during the dynamics and insensitive to 2 ms input noise. A decision is the first time $r_i \geq \theta$, with $\theta = 0.75\,r^*$ and $r^* = 30.1$ Hz the winning-attractor rate at baseline ($\theta = 22.6$ Hz). The threshold is calibrated once at baseline and kept fixed across sweeps, treating it as a fixed measurement device. Trials are *correct* or *error* (population 1 or 2 crosses first), *timeout* (no crossing in 2 s), or *premature* (crossing before stimulus onset; excluded from RT/accuracy). Sensitivity to the fraction (0.5–0.9) is reported.

Deterministic analysis. Fixed points are found by vectorized damped Newton iteration from a grid of starts with the analytic Jacobian and classified by eigenvalues. The separatrix is the stable

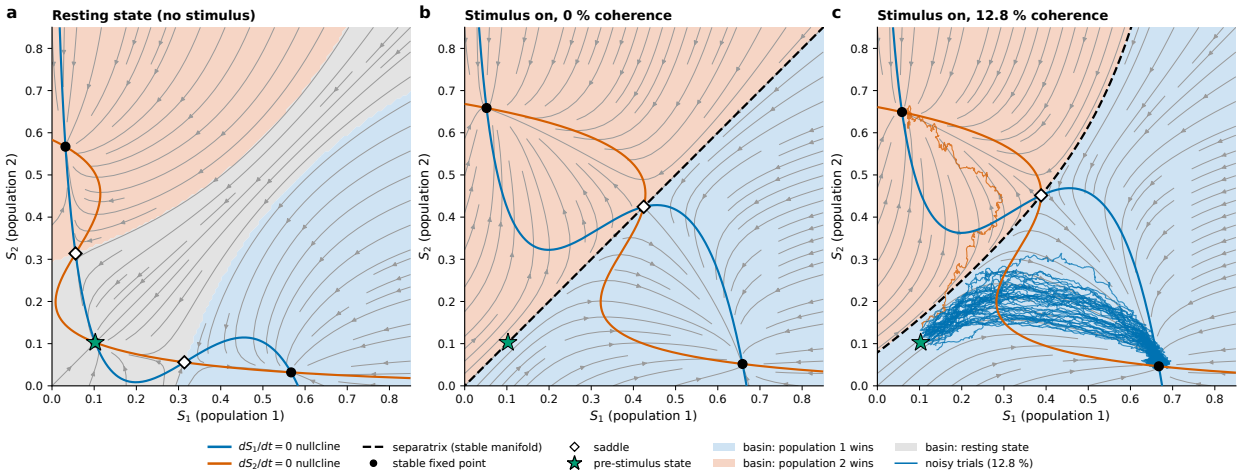


Figure 1: Phase plane of the reduced model. (a) Without stimulus: resting state (star), two persistent states, two saddles. (b) Stimulus at 0% coherence: the resting state has vanished and the symmetric saddle's stable manifold (the diagonal) separates two choice attractors. (c) At 12.8% coherence the saddle and separatrix shift; 40 noisy trials leave the resting state and nearly all fall into the correct basin, one error trial crossing the separatrix.

manifold of the decision saddle, obtained by integrating the reversed flow from the saddle along the stable eigenvector. d is the signed distance (in S units) from the resting state to the separatrix (positive if the deterministic outcome is correct). Bifurcation values are located by bisection on changes in the number of stable fixed points and saddles. Basins are obtained by integrating the deterministic flow from a 130×130 grid.

Experiments. Baseline: six coherences (0, 3.2, 6.4, 12.8, 25.6, 51.2%), 1000 trials each. Main sweep: $J_{12} \in [0.015, 0.085]$ nA $\times$ $\sigma \in [0.004, 0.08]$ nA, 15×15 grid, 500 trials per cell at 0% and 12.8% coherence. Supplementary sweep over (J_{11}, σ) with the same σ range.

Operational regimes (12.8% coherence). *Indecisive*: timeout rate $> 20\%$. *Impulsive*: (not indecisive and) accuracy < 0.65 or premature rate $> 10\%$. *Other*: mean RT at 12.8% not below RT at 0%. *Normal*: all else. The cut-offs were chosen from exploratory runs and then held fixed for all maps; the dependence of the result on the decision threshold is shown in Section 4.7.

Statistics. H1: ordinary least squares on log mean RT, and binomial (logistic) regression on error counts (McFadden pseudo-R^2), over cells with a saddle, ≥ 50 decided trials, and $< 5\%$ premature and timeout trials ($n = 171$). Predictors are standardized; generalization is assessed by 5-fold cross-validation that holds out whole J_{12} values, because geometric quantities depend only on J_{12} and cells sharing J_{12} are not independent. Binomial confidence intervals are Wilson intervals.

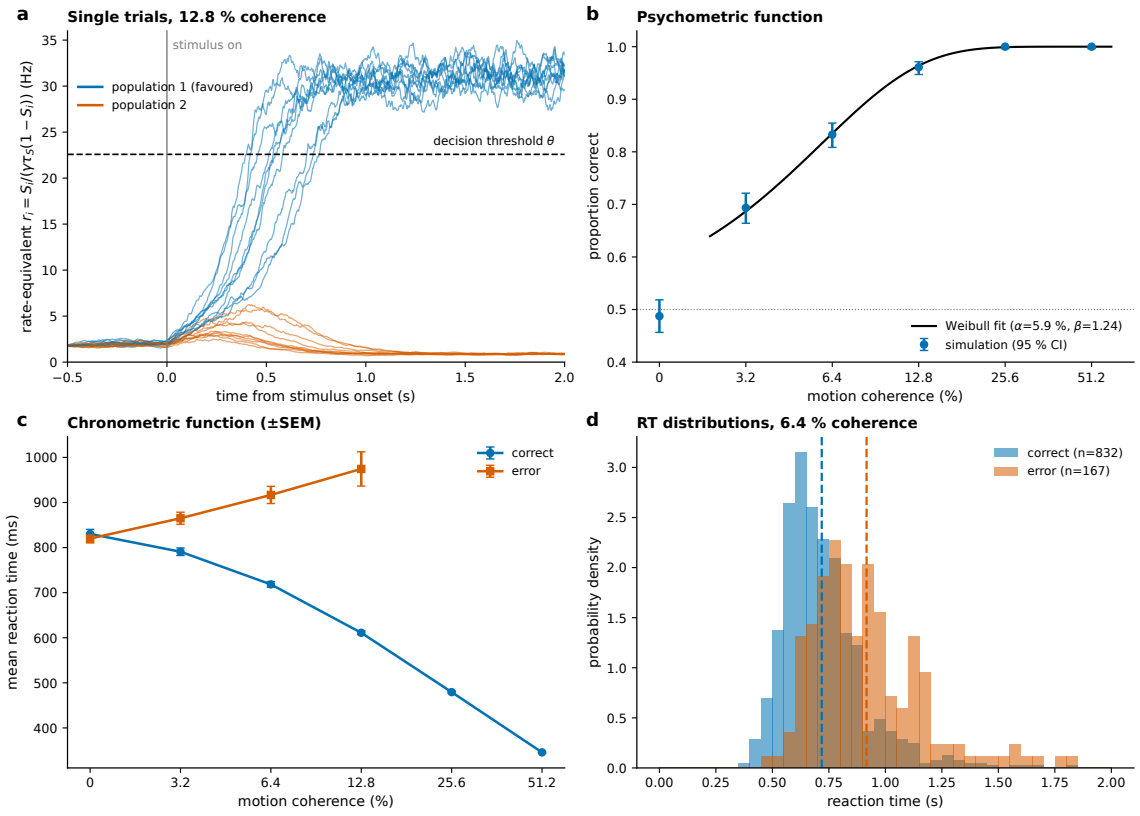


Figure 2: Baseline behavior ($\sigma = 0.02$ nA, 1000 trials per coherence). (a) Rate-equivalent traces. (b) Psychometric function with Weibull fit ($\alpha = 5.9\%$, $\beta = 1.24$). (c) Chronometric function: errors are slower than correct responses. (d) RT distributions at 6.4% coherence.

4 Results

4.1 Baseline behavior

Figure 1 shows the geometry. At baseline the stimulus-on system has two stable choice attractors (winner rate 30.1 Hz, loser < 1 Hz) and a saddle at $S_1 = S_2 \approx 0.42$ (≈ 11.5 Hz), as in the original paper. Single trials (Fig. 2a) show slow ramping over hundreds of ms through the saddle region followed by a commitment to one attractor.

Accuracy increases monotonically with coherence, RT falls from about 830 ms to 346 ms, and errors are slower than correct responses at all coherences where they occur (Table 1, Fig. 2). These qualitative signatures match Wong and Wang (2006) and the neural and behavioral findings in Roitman and Shadlen (2002). Accuracy and mean RT are unchanged within sampling error across the four integration steps (Fig. 8d–e).

4.2 Bifurcation structure

With the stimulus on at 0% coherence, the system has a single symmetric (co-activation) stable state for $J_{12} < 0.0247$ nA. At $J_{12} = 0.0247$ two mirror-image saddle-node bifurcations create the two choice attractors, and at $J_{12} = 0.0390$ the symmetric state loses stability (subcritical pitchfork,

Coherence (%)	Accuracy	RT correct (ms)	RT error (ms)	Timeouts (%)
0	0.487	830	820	0.5
3.2	0.693	791	865	0.5
6.4	0.833	718	917	0.1
12.8	0.961	611	974	0.1
25.6	1.000	479	–	0.0
51.2	1.000	346	–	0.0

Table 1: Baseline summary (RT error shown only if ≥ 5 errors). At 0% coherence "correct" is arbitrary.

	log RT (OLS)		error rate (logistic)	
Predictors	R^2	CV R^2	pseudo-R^2	CV
σ	0.57	0.56	0.81	0.81
J_{12}, σ	0.83	0.81	0.81	0.81
$\log \lambda_u, \sigma$	0.78	0.75	0.81	0.81
$\log \lambda_u, d, \sigma$	0.80	0.56	0.81	0.80
$\log \lambda_u, d, d_{\mathrm{sad}}, \sigma$	0.84	0.56	0.81	0.80

Table 2: H1 model comparison over $n = 171$ cells. CV holds out whole J_{12} values.

becoming the decision saddle). At 12.8% coherence the same sequence is split into three events at 0.0212, 0.0283 and 0.0318 (Fig. 3). The baseline $J_{12} = 0.0497$ lies in the winner-take-all range. Along the J_{11} axis the choice attractors exist only above $J_{11} = 0.2191$ nA (0% coherence).

4.3 Regime map

Out of $15^2 = 225$ cells, 144 are normal and 70 impulsive; no cell is indecisive (Fig. 4e). Normal behavior holds below $\sigma \approx 0.05$ nA over the whole J_{12} range. Higher noise yields evidence-insensitive choices (accuracy near 0.6) with shortened RT, plus a growing fraction of responses *before* the stimulus at small J_{12} (Fig. 4c). RT increases steadily with J_{12} (stronger inhibition slows the escape), and no timeouts appear even for the strongest inhibition (RT saturates near 1 s for J_{12} up to 0.4 nA in exploratory runs). Note that the operational "normal" label also covers $J_{12} < 0.0212$ nA, where a co-activation state replaces winner-take-all competition: there the losing population reaches 10–15 Hz at the decision (Fig. 4d) while accuracy remains high at low noise. Accuracy and RT alone therefore cannot detect the loss of competition, a limitation of the operational definitions.

4.4 H1: geometric predictors of RT and errors

The unstable eigenvalue λ_u varies non-monotonically with J_{12} (maximum near 0.055 nA), while d is almost constant (≈ 0.045) once $J_{12} \gtrsim 0.045$ nA and only departs from it near the saddle-node (Fig. 5a). For log RT, σ alone gives $R^2 = 0.57$; adding $\log \lambda_u$ raises it to 0.78 (cross-validated 0.75), and removing λ_u from the three-predictor model costs 0.22 R^2, the largest geometric effect (Table 2). Adding d gains only about 0.02 in-sample and *reduces* cross-validated R^2 to 0.56, because a few cells near the saddle-node have extreme d and high leverage. The parameter model $\{J_{12}, \sigma\}$ is as good as or better (0.83, CV 0.81), so λ_u adds interpretation but no predictive power beyond

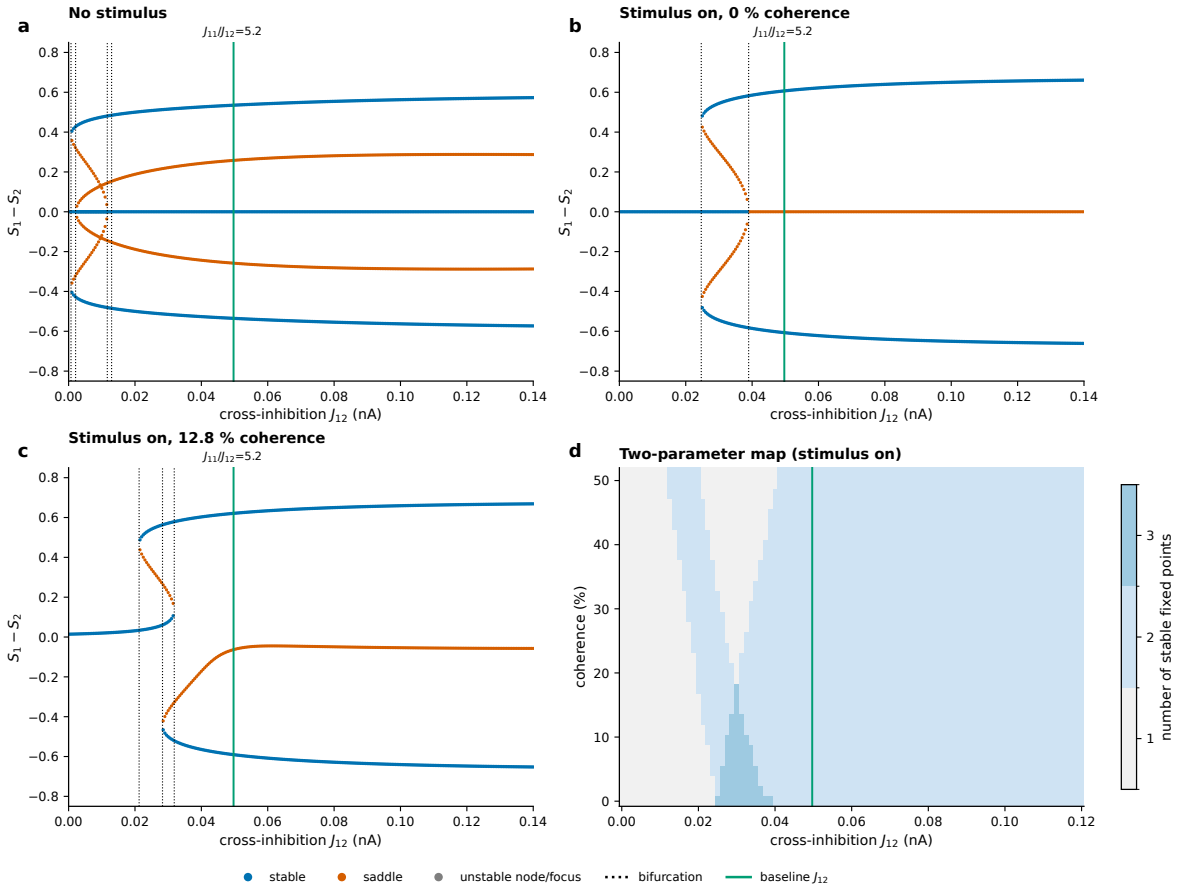


Figure 3: Fixed points versus cross-inhibition J_{12} (order parameter $S_1 - S_2$). (a) No stimulus. (b) Stimulus at 0% coherence. (c) At 12.8%, where the symmetry is broken. (d) Number of stable states over the (J_{12}, coherence) plane. The green line marks the baseline J_{12}.

J_{12} itself; with a single swept circuit parameter, geometry cannot be separated from any monotone function of J_{12}. For the error rate at 12.8%, σ alone gives pseudo-$R^2 = 0.81$ and the geometric terms add < 0.003. **H1 is therefore partly supported:** λ_u explains speed, d is not informative here, and the error rate is a noise effect.

4.5 H2: regime boundaries versus bifurcations

In the (J_{12}, σ) plane deterministic bifurcations are vertical lines (Fig. 4e). The σ at which accuracy reaches 0.65 varies between 0.048 and 0.066 nA with only a weak, non-monotonic dependence on J_{12} (Spearman $\rho = 0.33$, $p = 0.23$) and shows no step or kink at the bifurcation lines; the losing-population rate changes smoothly through them (Fig. 4d). The premature-response boundary, however, moves from 0.061 to 0.080 nA as J_{12} increases ($\rho = 0.98$, $n = 9$ cells with a crossing) and is linearly predicted by a deterministic quantity, the distance between the resting state and the nearest saddle without stimulus ($R^2 = 0.98$, Fig. 4f): a larger barrier tolerates more noise. Thus noise-driven boundaries are organized by the deterministic geometry, but *not* by the bifurcation values themselves.

The indecisive regime appears only when J_{11} is reduced (Fig. 6). Its boundary lies at $J_{11} =$

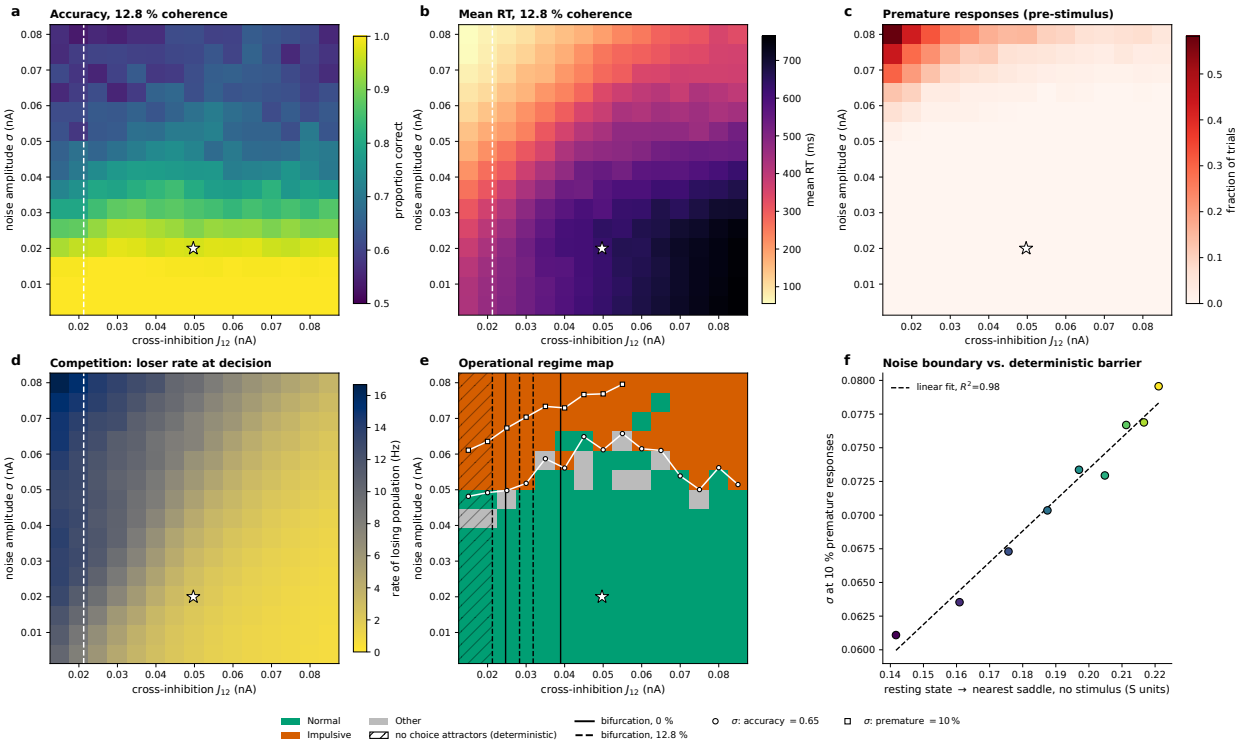


Figure 4: Behavior over (J_{12}, σ) at 12.8% coherence (star: baseline). (a) Accuracy. (b) Mean RT. (c) Premature responses. (d) Rate of the losing population at the decision. (e) Operational regime map with the deterministic bifurcation values (solid: 0%; dashed: 12.8%), the hatched region without choice attractors, and the σ at which accuracy reaches 0.65 or premature responses reach 10%. (f) The premature-response boundary against the distance from the resting state to its nearest saddle (no stimulus).

0.231–0.239 nA, *above* the existence bifurcation of the choice attractors (0.2191–0.2223 nA) and just below the value 0.241 nA at which the attractor rate drops to the criterion θ. Noise shifts the boundary in the predicted direction (toward weaker J_{11}, i.e. noise-assisted decisions; slope -0.05 nA/nA, $R^2 = 0.35$), but weakly. Hence H2 holds only in this weaker sense: the boundary follows a deterministic curve shifted by noise, but that curve is set partly by the read-out criterion, not by a bifurcation. For large J_{11} and high noise, impulsive behavior reappears.

4.6 Empirical landscape

The system is non-gradient, so $-\log P$ is only an empirical landscape. Long simulations (20000 s each) show a double well at low noise with rare transitions (4 at $\sigma = 0.05$, 173 at 0.055, 963 at 0.06 nA, counting changes between choice states), and a single broad well at 0.07 nA where the barrier has collapsed (Fig. 7). This collapse occurs in the range where accuracy at 12.8% falls below 0.65, consistent with noise-induced loss of the choice structure underlying the impulsive regime. This is a correlation across noise levels, not a causal test.

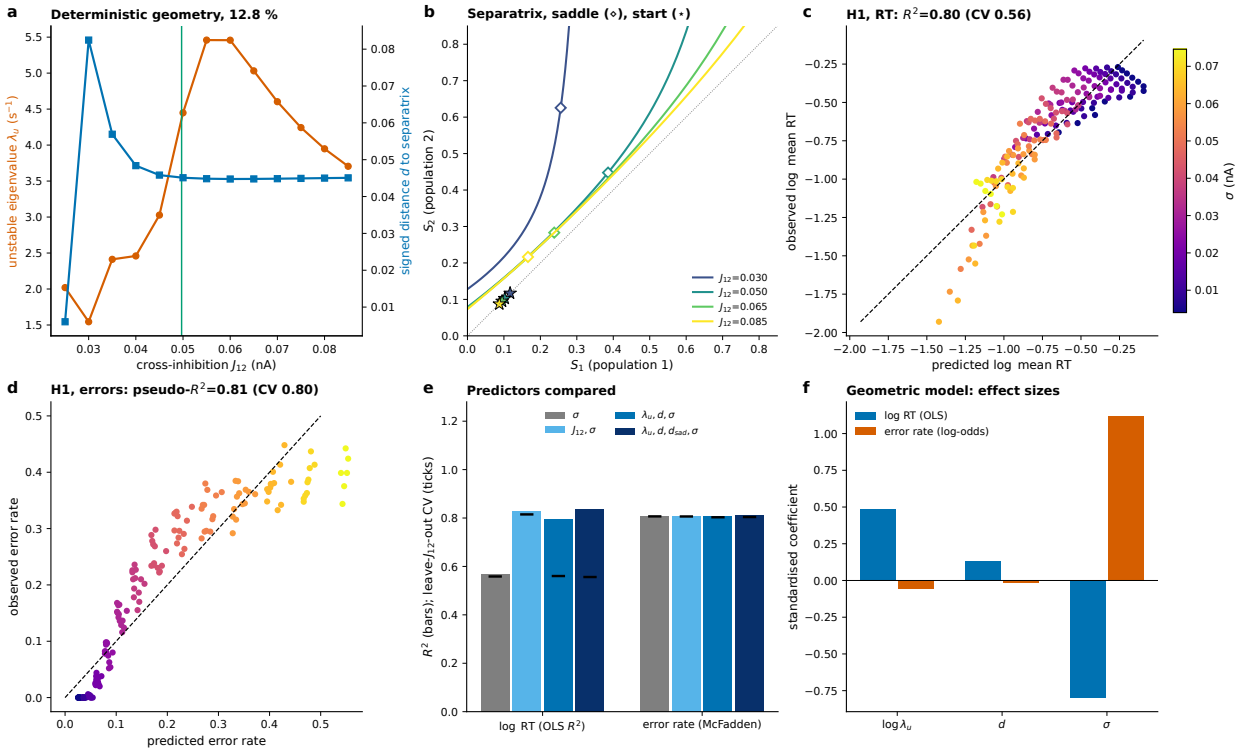


Figure 5: (a) λ_u and d versus J_{12} at 12.8%. (b) Separatrices, decision saddles ($\diamond$) and resting states ($\star$). (c,d) Predicted versus observed log RT and error rate for the model $\{\lambda_u, d, \sigma\}$. (e) R^2 (bars) and cross-validated R^2 (ticks) of competing models. (f) Standardized coefficients.

4.7 Sensitivity

Accuracy is independent of the threshold fraction (the winner is determined early), while mean RT shifts monotonically (e.g. about 660 to 960 ms at 0% between $f = 0.5$ and 0.9) (Fig. 8). The area of the impulsive regime does depend on f: 85, 70 and 39 of 225 cells for $f = 0.6$, 0.75 and 0.9. The qualitative picture is stable but the quantitative position of the boundary is criterion-dependent.

5 Discussion and Limitations

Three results stand out. First, behavior depends smoothly on J_{12}: deterministic bifurcations (appearance of choice attractors, pitchfork) are not visible as discrete behavioral transitions in accuracy or RT. Second, the dominant control of dysfunctional behavior is noise, which both produces premature responses (through escape from the resting state, with a tolerance set by the barrier) and merges the choice states (impulsive regime). Third, the indecisive regime requires weakening recurrent excitation, not strengthening inhibition, so the effective ratio J_{11}/J_{12} should be interpreted with both terms in mind.

Limitations. (i) Regime definitions are operational, evaluated only at 12.8% coherence, and criterion-dependent (Section 4.7). (ii) The read-out threshold is fixed across the sweeps, so part of the indecisive boundary reflects the read-out. (iii) Geometric predictors vary along a single pa-

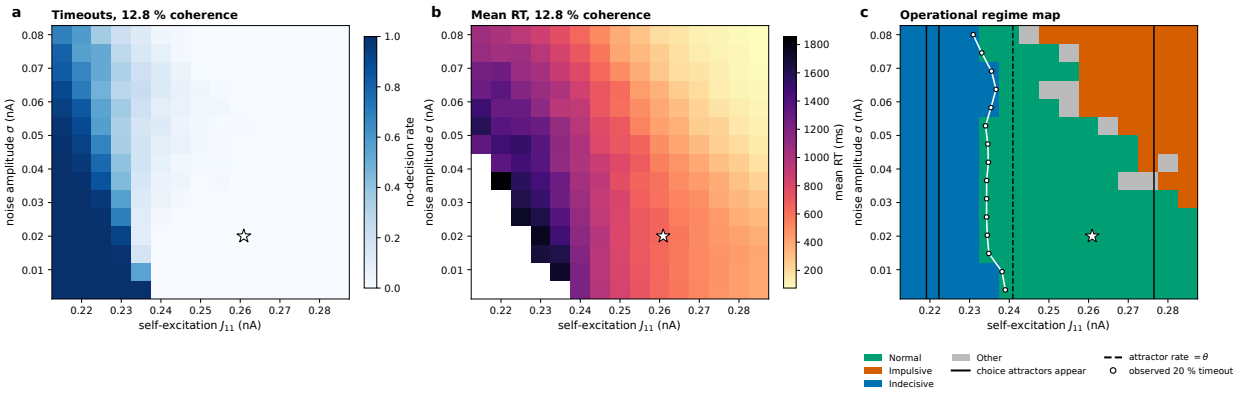


Figure 6: Supplementary sweep over self-excitation J_{11} and σ (12.8% coherence). (a) Timeout rate. (b) Mean RT (white: no decisions). (c) Regime map with the saddle-node values (solid), the J_{11} at which the attractor rate equals θ (dashed), and the observed 20% timeout boundary.

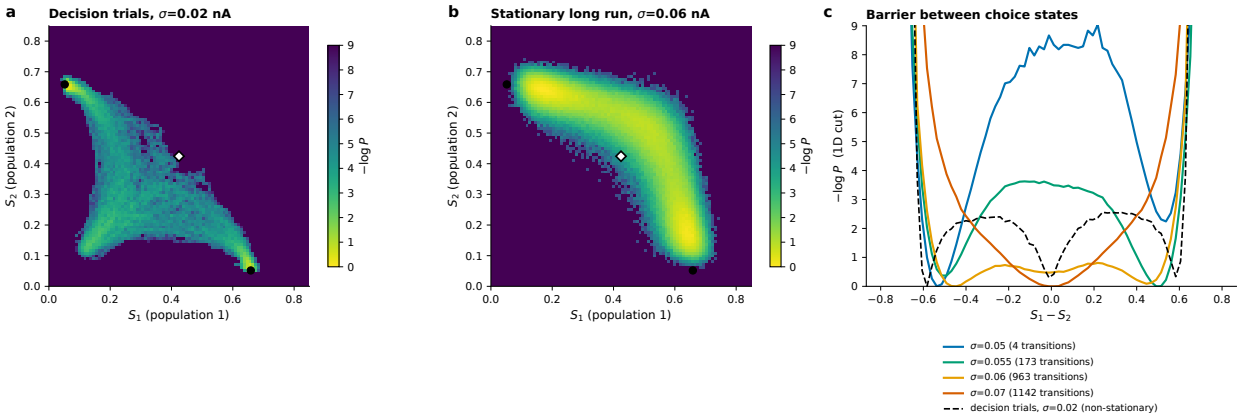


Figure 7: Empirical quasi-potential $-\log P$ at 0% coherence. (a) Occupancy of decision trials at baseline noise (non-stationary). (b) Stationary long run at $\sigma = 0.06$ nA. (c) 1D cut along $S_1 - S_2$ for several σ.

rameter (J_{12}), which limits H1. (iv) Statistics treat cells as independent apart from the grouped cross-validation. (v) The model is a mean-field reduction fitted to nothing; no claim is made about biological data or clinical conditions. (vi) Parameter values were not verified against the original paper's table. (vii) The 12.8% condition has few errors at low noise, limiting error-rate analysis.

Deviations from the proposal. OU noise is advanced exactly rather than by Euler–Maruyama; the read-out uses the S-based rate; a premature-response category and a supplementary J_{11} sweep were added; the Lam et al. (2022) reference was removed because its details could not be verified; the drift–diffusion fit, perseveration protocol and spiking-network cross-check remain out of scope.

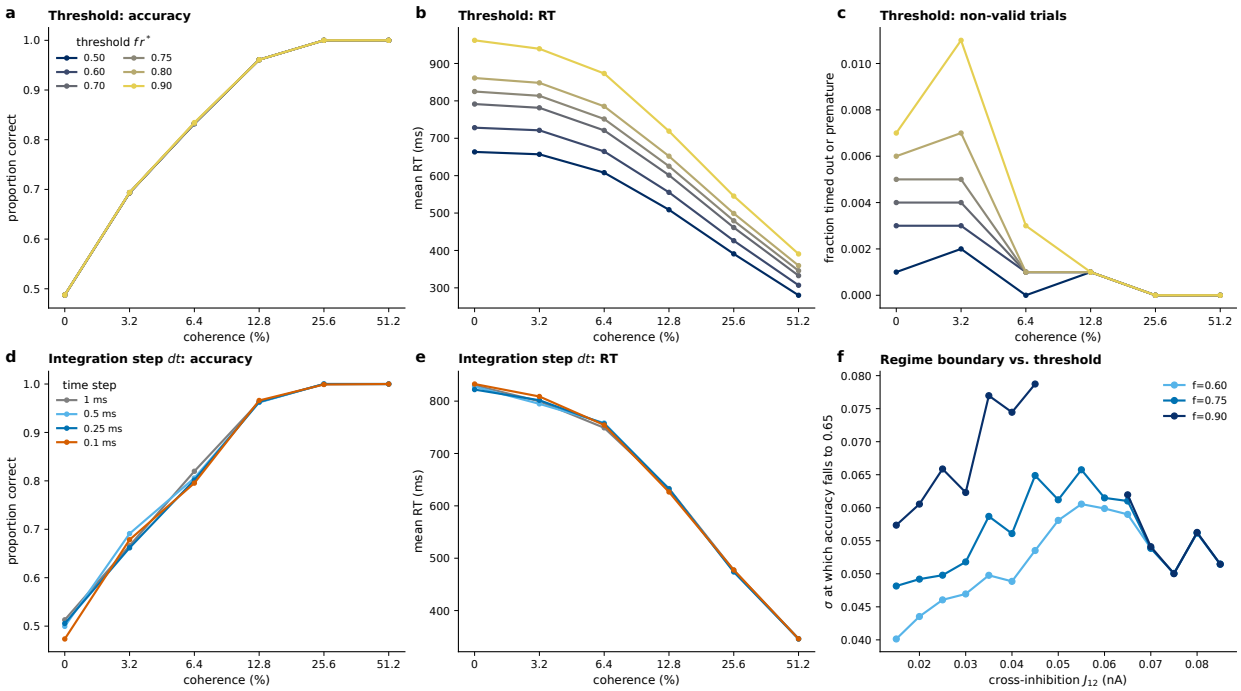


Figure 8: Sensitivity. (a–c) Threshold fraction f (0.5–0.9). (d,e) Integration step. (f) Accuracy boundary versus J_{12} for three thresholds.

6 Conclusion

The reduced Wong–Wang circuit reproduces classic behavior and has an interpretable geometry. Speed is governed by the saddle's unstable eigenvalue, accuracy by noise relative to a deterministic barrier, and dysfunctional regimes by noise (impulsive) or weak recurrent excitation (indecisive). Deterministic bifurcations organize the phase plane but, in this read-out, do not align with behavioral regime boundaries.

A Parameters

Symbol	Value	Meaning
τ_S	100 ms	NMDA time constant
γ	0.641	gating kinetics
a, b, d	270 Hz/nA, 108 Hz, 0.154 s	input–output function
J_{11}	0.2609 nA	self-excitation
J_{12}	0.0497 nA	cross-inhibition
I_0	0.3255 nA	background input
J_{ext}, μ_0	5.2×10^{-4} nA/Hz, 30 Hz	stimulus coupling and strength
σ, τ_{AMPA}	0.02 nA, 2 ms	noise

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